Chapter 8. Universal Approximation, Barron's Theorem and Generalization

Lectures 10/21, 10/26: measures, normed spaces, Cybenko's theorem; Barron's theorem and proof; Rademacher complexity

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1 Three Sources of Error

Let f^* be the ideal target, $\mathcal{F}$ the class of functions a given architecture can represent, f_{app} the best element of $\mathcal{F}$, f_{gen} the best element on the finite training sample (empirical risk minimizer) and f_{opt} what the optimizer actually returns. The total error splits as

$$\underbrace{f^* \to f_{\text{app}}}_{\text{approximation}} + \underbrace{f_{\text{app}} \to f_{\text{gen}}}_{\text{generalization (estimation)}} + \underbrace{f_{\text{gen}} \to f_{\text{opt}}}_{\text{optimization}}.$$

Chapter 6 dealt with optimization; this chapter shows that the approximation error of one-hidden-layer networks can be made arbitrarily small (universality) and, for Barron functions, quantifies it.

2 Preliminaries from Analysis

2.1 Measures

Definition 8.1: measures on a compact set $K \subset \mathbb{R}^n$

The **Borel sets** of K form the smallest collection containing all open subsets of K and closed under complements in K and countable unions. A (nonnegative) **measure** μ assigns $\mu(A) \in [0, \infty]$ to Borel sets with

$$\mu(\emptyset) = 0, \qquad \mu\Big(\bigcup_{j=1}^{\infty} A_j\Big) = \sum_{j=1}^{\infty} \mu(A_j) \quad \text{for pairwise disjoint } A_j.$$

A **signed measure** is the same with real (possibly negative) values, e.g. a difference of two finite measures. $\mu = 0$ means $\mu(A) = 0$ for every Borel set A. $M(I_n)$ denotes the finite signed regular Borel measures on the unit cube $I_n = [0, 1]^n$.

2.2 Normed spaces and density

Definition 8.2: linear and normed spaces

A real **linear space** is a set closed under addition and multiplication by reals obeying the usual vector-space rules; e.g. $C(K) = \{f : K \to \mathbb{R} \text{ continuous}\}$, since $af + bg \in C(K)$ with $(af + bg)(x) = af(x) + bg(x)$. A **normed linear space** is a linear space X with $\|\cdot\| : X \to [0, \infty)$ such that

$$\|u\| = 0 \iff u = 0, \qquad \|cu\| = |c|\|u\|, \qquad \|u + v\| \le \|u\| + \|v\|.$$

On $C(K)$ we use the sup norm $\|f\|_\infty = \max_{x \in K} |f(x)|$. A subset $D \subseteq X$ is **dense** in X if for every $f \in X$ and $\varepsilon > 0$ there is $g \in D$ with $\|f - g\| < \varepsilon$.

Density in $(C(I_n), \|\cdot\|_\infty)$ is exactly uniform approximation: every continuous function can be approximated within ε at every point simultaneously.

3 Cybenko's Universal Approximation Theorem

Definition 8.3: sigmoidal and discriminatory

$\sigma : \mathbb{R} \to [0,1]$ is **sigmoidal** if $\lim_{t\to-\infty}\sigma(t) = 0$ and $\lim_{t\to+\infty}\sigma(t) = 1$. For $\mu \in M(I_n)$, σ is **discriminatory** if

$$\int_{I_n} \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu(\mathbf{x}) = 0 \quad \text{for all } \mathbf{w} \in \mathbb{R}^n,\ \theta \in \mathbb{R} \qquad \Longrightarrow \qquad \mu = 0.$$

Theorem 8.4: Cybenko 1989 (UAT)

Let σ be any continuous sigmoidal function. Then finite sums

$$G(\mathbf{x}) = \sum_{j=1}^{N} \alpha_j\, \sigma(\mathbf{w}_j^\top \mathbf{x} + \theta_j), \qquad \mathbf{w}_j \in \mathbb{R}^n,\ \alpha_j, \theta_j \in \mathbb{R},\ N \in \mathbb{N},$$

are dense in $C(I_n)$: for every continuous f and $\varepsilon > 0$ some G satisfies $\sup_{\mathbf{x}\in I_n} |f(\mathbf{x}) - G(\mathbf{x})| < \varepsilon$.

The proof combines three ingredients: a Hahn–Banach consequence (Theorem 8.5), the Riesz representation theorem (Theorem 8.6), and the discriminatory property (Theorem 8.9).

3.1 The functional-analytic lemmas

Theorem 8.5: a consequence of Hahn–Banach

Let W be a linear, non-dense subspace of a normed linear space X. Then there is a bounded linear functional L on X with $L \neq 0$ on X and $L|_W = 0$.

Proof. The closure $\overline{W}$ is a closed subspace and $\overline{W} \neq X$, so pick $x_0 \notin \overline{W}$; then $d = \inf_{w\in\overline{W}} \|x_0 - w\| > 0$. On $\overline{W} \oplus \mathbb{R}x_0$ define $\ell(w + tx_0) = t$. It is linear, vanishes on W, $\ell(x_0) = 1$, and bounded: $|t|d \le |t|\|x_0 + w/t\| = \|w + tx_0\|$ for $t \neq 0$, so $\|\ell\| \le 1/d$. The Hahn–Banach theorem extends ℓ to X with the same norm. ∎

Theorem 8.6: Riesz representation theorem

Let $K \subset \mathbb{R}^n$ be compact. For every bounded linear functional F on $C(K)$ there is a unique finite signed (regular Borel) measure μ on K with

$$F(f) = \int_K f(\mathbf{x})\, d\mu(\mathbf{x}) \qquad \text{for all } f \in C(K).$$

(For positive functionals the measure is nonnegative.)

Theorem 8.7: an annihilating measure

Let W be a linear, non-dense subspace of $C(I_n)$. Then there is $\mu \in M(I_n)$, $\mu \neq 0$, with $\int_{I_n} h\, d\mu = 0$ for all $h \in W$.

Proof. Theorem 8.5 with $X = C(I_n)$ gives a bounded linear $L \neq 0$ with $L|_W = 0$. Riesz represents $L(f) = \int f\, d\mu$; $\mu \neq 0$ because $L \neq 0$; and for $h \in W$, $\int h\, d\mu = L(h) = 0$. ∎

3.2 Density for discriminatory functions

Theorem 8.8: density for discriminatory activations

If σ is continuous and discriminatory, the sums G above are dense in $C(I_n)$.

Proof. Since σ is continuous, $W = \{G\}$ is a linear subspace of $C(I_n)$ (sums and scalar multiples of such sums are again such sums). Suppose W is not dense. By Theorem 8.7 there is $\mu \neq 0$ with $\int h\, d\mu = 0$ for all $h \in W$, in particular

$$\int_{I_n} \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu(\mathbf{x}) = 0 \qquad \text{for all } \mathbf{w} \in \mathbb{R}^n,\ \theta \in \mathbb{R}.$$

Because σ is discriminatory, $\mu = 0$: a contradiction. ■

3.3 Continuous sigmoidals are discriminatory

Theorem 8.9: continuous sigmoidals are discriminatory

Every continuous sigmoidal function is discriminatory for all $\mu \in M(I_n)$.

Proof. Assume $\int \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu = 0$ for all $\mathbf{w}, \theta$. Fix $\mathbf{w}, \theta, \varphi$ and let $\sigma_\lambda(\mathbf{x}) = \sigma\big(\lambda(\mathbf{w}^\top \mathbf{x} + \theta) + \varphi\big)$. As $\lambda \to \infty$, $\sigma_\lambda \to 1$ on the open half-space $\Pi = \{\mathbf{w}^\top \mathbf{x} + \theta > 0\}$, $\to 0$ on $\{\mathbf{w}^\top \mathbf{x} + \theta < 0\}$ and $= \sigma(\varphi)$ on the hyperplane $H = \{\mathbf{w}^\top \mathbf{x} + \theta = 0\}$. The σ_λ are bounded by 1, so dominated convergence gives

$$0 = \int \sigma_\lambda\, d\mu \ \longrightarrow\ \mu(\Pi) + \sigma(\varphi)\mu(H) \qquad \text{for every } \varphi.$$

Letting $\varphi \to -\infty$ and $\varphi \to +\infty$ yields $\mu(\Pi) = 0$ and $\mu(H) = 0$ for all half-spaces and hyperplanes. Fix $\mathbf{w}$ and define $F(h) = \int h(\mathbf{w}^\top \mathbf{x})\, d\mu(\mathbf{x})$ for bounded measurable $h : \mathbb{R} \to \mathbb{R}$. F is linear and bounded, and $F(\mathbf{1}_{[\theta,\infty)}) = \mu(\Pi) + \mu(H) = 0$, $F(\mathbf{1}_{(\theta,\infty)}) = 0$, hence F vanishes on indicators of intervals, on simple functions, and by uniform approximation on all bounded measurable h. Taking $h = \cos$ and $h = \sin$ gives $\int e^{i\mathbf{w}^\top \mathbf{x}}\, d\mu(\mathbf{x}) = 0$ for every $\mathbf{w}$: the Fourier transform of μ vanishes, so $\mu = 0$. ■

Proof of Cybenko's theorem. Theorem 8.9 shows σ is discriminatory; Theorem 8.8 gives density. ■

Remark 8.10: what the theorem says and does not say

One hidden layer with enough neurons approximates any continuous function on a compact set uniformly (the cube can be replaced by any compact K after rescaling). The theorem is an existence result: it gives no bound on N, no algorithm to find the weights, and no statement about generalization. Later work extends it to every continuous non-polynomial activation, including ReLU (Leshno et al. 1993). Intuition in 1D: a steep sigmoid is almost a step, a difference of two steps is a bump, and a sum of bumps approximates any continuous function.

4 Barron's Theorem: a Dimension-Free Rate

4.1 Fourier preliminaries

For $f \in L^1(\mathbb{R}^d)$ the Fourier transform is $\hat{f}(\boldsymbol{\xi}) = \int_{\mathbb{R}^d} f(\mathbf{x}) e^{-2\pi i \boldsymbol{\xi}^\top \mathbf{x}}\, d\mathbf{x}$. The Sobolev space $H^s(\mathbb{R}^d)$ consists of f with $\int |\hat{f}(\boldsymbol{\xi})|^2 (1 + |\boldsymbol{\xi}|^2)^s\, d\boldsymbol{\xi} < \infty$.

Definition 8.11: Barron class

For $C > 0$,

$$\Gamma_C = \Big\{ f \in L^1(\mathbb{R}^d) : \ \|\hat{f}\|_{L^1(\mathbb{R}^d)} < \infty, \ \int_{\mathbb{R}^d} |2\pi\boldsymbol{\xi}| \, |\hat{f}(\boldsymbol{\xi})| \, d\boldsymbol{\xi} < C \Big\}.$$

On a domain Ω one also uses the spectral Barron norm $\|f\|_{\mathcal{B}^s(\Omega)} = \inf_{f_e|_\Omega = f} \int_{\mathbb{R}^d} (1 + |\boldsymbol{\xi}|)^s |\hat{f}_e(\boldsymbol{\xi})| \, d\boldsymbol{\xi}$, the infimum over all extensions f_e of f beyond Ω.

Theorem 8.12 (Theorem 1, Barron 1993)

Let σ be sigmoidal, $f \in \Gamma_C$ and $B_1^d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \le 1\}$. Then for every $c > 4C^2$ and every $N \in \mathbb{N}$ there is a network f_N with N neurons such that

$$\frac{1}{|B_1^d|} \int_{B_1^d} |f(\mathbf{x}) - f_N(\mathbf{x})|^2 \, d\mathbf{x} < \frac{c}{N}.$$

The rate $O(1/N)$ does not depend on the dimension d. The price is the assumption $f \in \Gamma_C$, a condition on the Fourier spectrum.

Remark 8.13: curse of dimensionality

Approximating a general function with s bounded derivatives in d variables to accuracy ε by a fixed linear family needs on the order of $\varepsilon^{-d/s}$ terms, which explodes with d. Barron's bound avoids this: the number of neurons needed for mean squared error ε is about c/ε for every d. The dimension enters only through the constant C.

5 Proof of Barron's Theorem

The proof (lecture 10/26) has three steps. Lemma 1 (Maurey) says that an element of a closed convex hull is approximated at rate $1/N$ by averages of N elements. Lemma 2 places every Barron function in such a hull, namely the hull of scaled half-space indicators. A final step replaces the indicators by sigmoids.

5.1 Convex hulls and Maurey's lemma

Definition 8.14: convex hull

For a subset G of a vector space,

$$\operatorname{co}(G) = \Big\{ \sum_{j=1}^{n} \alpha_j x_j : \ n \in \mathbb{N}, \ x_j \in G, \ \alpha_j \ge 0, \ \sum_{j=1}^{n} \alpha_j = 1 \Big\},$$

and $\overline{\operatorname{co}}(G)$ denotes its closure in a given norm.

Theorem 8.15 (Lemma 1, Maurey)

Let H be a Hilbert space and $G \subset H$ with $\|g\|_H \le B$ for all $g \in G$, for some $B > 0$. Let $f \in \overline{\mathrm{co}}(G)$. Then for every $N \in \mathbb{N}$ and every $c > B^2$ there exist $g_1, \dots, g_N \in G$ with

$$\Big\| f - \frac{1}{N} \sum_{i=1}^{N} g_i \Big\|_H^2 \le \frac{c}{N}.$$

Proof. Fix N and $\varepsilon > 0$. Since $f \in \overline{\mathrm{co}}(G)$ there are $\alpha_1, \dots, \alpha_m \in [0,1]$ summing to 1 and $h_1, \dots, h_m \in G$ such that

$$f^* = \sum_{j=1}^{m} \alpha_j h_j \quad \text{satisfies} \quad \|f - f^*\|_H < \varepsilon.$$

Claim. There are $g_1, \dots, g_N \in \{h_1, \dots, h_m\}$ with $\big\| f^* - \frac{1}{N} \sum_i g_i \big\|_H^2 \le B^2/N$.

Let $X_1, \dots, X_N$ be iid random elements with $P(X_i = h_j) = \alpha_j$. Then $\mathbb{E}[X_i] = \sum_j P(X_i = h_j)\, h_j = f^*$. Writing $f^* = \frac{1}{N} \sum_{j=1}^{N} f^*$ and using $\|\sum_j v_j\|^2 = \sum_j \|v_j\|^2 + \sum_{i \ne j} \langle v_i, v_j \rangle$,

$$\mathbb{E}\Big\| f^* - \frac{1}{N} \sum_{j=1}^{N} X_j \Big\|^2 = \frac{1}{N^2}\, \mathbb{E}\Big[\sum_{j=1}^{N} \|f^* - X_j\|^2 + \sum_{i \ne j} \langle f^* - X_i, f^* - X_j \rangle \Big].$$

For $i \ne j$, independence gives $\mathbb{E}\langle X_i, X_j \rangle = \langle \mathbb{E}X_i, \mathbb{E}X_j \rangle = \langle f^*, f^* \rangle$, so

$$\mathbb{E}\langle f^* - X_i, f^* - X_j \rangle = \langle f^*, f^* \rangle - \langle f^*, \mathbb{E}X_j \rangle - \langle \mathbb{E}X_i, f^* \rangle + \mathbb{E}\langle X_i, X_j \rangle = 0.$$

The N diagonal terms have the same distribution, hence

$$\begin{aligned} \mathbb{E}\Big\| f^* - \frac{1}{N} \sum_{j} X_j \Big\|^2 &= \frac{1}{N} \mathbb{E}\|f^* - X_1\|^2 = \frac{1}{N} \mathbb{E}[\|f^*\|^2 - 2\langle f^*, X_1 \rangle + \|X_1\|^2] \\ &= \frac{1}{N} (\mathbb{E}\|X_1\|^2 - \|f^*\|^2) \le \frac{B^2}{N}, \end{aligned}$$

using $\mathbb{E}\langle f^*, X_1 \rangle = \|f^*\|^2$ and $\|X_1\| \le B$. A random variable cannot exceed its mean for every outcome, so some realization $g_i = X_i(\omega) \in \{h_1, \dots, h_m\}$ satisfies the claim.

By the triangle inequality,

$$\Big\| f - \frac{1}{N} \sum_{i} g_i \Big\|_H \le \|f - f^*\|_H + \Big\| f^* - \frac{1}{N} \sum_{i} g_i \Big\|_H < \varepsilon + \frac{B}{\sqrt{N}}.$$

Since $c > B^2$, choosing $\varepsilon < (\sqrt{c} - B)/\sqrt{N}$ makes the right-hand side less than $\sqrt{c/N}$; squaring gives the lemma. ∎

This is what Theorem 1 needs: if f lies in the closed convex hull of a set of single neurons bounded by B in the normalized L^2 norm, an average of N neurons approximates it with error c/N. So it suffices to show that every $f \in \Gamma_C$ lies in the closed convex hull of affinely transformed Heaviside functions.

5.2 Barron functions lie in a convex hull of half-space indicators

On the unit ball use the normalized norm

$$\|g\|_{L_0^2(B_1^d)} = \Big(\frac{1}{|B_1^d|} \int_{B_1^d} |g(\mathbf{x})|^2 \, d\mathbf{x} \Big)^{1/2},$$

and the set of scaled half-space indicators

$$G_C = \Big\{ \mathbf{x} \in B_1^d \mapsto \gamma\, \mathbf{1}_{\mathbb{R}_+}(\langle \mathbf{a}, \mathbf{x} \rangle + b) : \ \mathbf{a} \in \mathbb{R}^d,\ b \in \mathbb{R},\ |\gamma| \le 2C \Big\},$$

where $\mathbf{1}_{\mathbb{R}_+}(t) = 1$ for $t \ge 0$ and 0 otherwise (the Heaviside function). Every $g \in G_C$ has $\|g\|_{L_0^2(B_1^d)} \le |\gamma| \le 2C$.

Theorem 8.16 (Lemma 2, Barron 1993)

Let $d \in \mathbb{N}$, $C > 0$ and $f \in \Gamma_C$. Then

$$f|_{B_1^d} - f(\mathbf{0}) \in \overline{\mathrm{co}}(G_C),$$

where the closure is taken in $L_0^2(B_1^d)$.

Sketch of proof. Since $f \in \Gamma_C$, both f and $\hat{f}$ are in $L^1(\mathbb{R}^d)$, so Fourier inversion holds:

$$f(\mathbf{x}) - f(\mathbf{0}) = \int_{\mathbb{R}^d} \hat{f}(\boldsymbol{\xi})(e^{2\pi i \langle \mathbf{x}, \boldsymbol{\xi} \rangle} - 1)\, d\boldsymbol{\xi}.$$

Write $\hat{f}(\boldsymbol{\xi}) = |\hat{f}(\boldsymbol{\xi})| e^{i\kappa(\boldsymbol{\xi})}$, where $\kappa(\boldsymbol{\xi})$ is the phase. Then

$$f(\mathbf{x}) - f(\mathbf{0}) = \int_{\mathbb{R}^d} |\hat{f}(\boldsymbol{\xi})| (e^{i(2\pi \langle \mathbf{x}, \boldsymbol{\xi} \rangle + \kappa(\boldsymbol{\xi}))} - e^{i\kappa(\boldsymbol{\xi})})\, d\boldsymbol{\xi}.$$

For real-valued f the left side is real, so we may take real parts:

$$f(\mathbf{x}) - f(\mathbf{0}) = \int_{\mathbb{R}^d} \frac{\cos\big(2\pi \langle \mathbf{x}, \boldsymbol{\xi} \rangle + \kappa(\boldsymbol{\xi})\big) - \cos \kappa(\boldsymbol{\xi})}{|2\pi \boldsymbol{\xi}|}\, |2\pi \boldsymbol{\xi}|\, |\hat{f}(\boldsymbol{\xi})|\, d\boldsymbol{\xi}.$$

Let $C_f = \int |2\pi \boldsymbol{\xi}| |\hat{f}(\boldsymbol{\xi})|\, d\boldsymbol{\xi} < C$ and define the probability measure Λ on $\mathbb{R}^d$ by $d\Lambda(\boldsymbol{\xi}) = C_f^{-1} |2\pi \boldsymbol{\xi}| |\hat{f}(\boldsymbol{\xi})|\, d\boldsymbol{\xi}$ (the lecture writes C for C_f). With

$$q_{\boldsymbol{\xi}}(\mathbf{x}) = \frac{\cos\big(2\pi \langle \mathbf{x}, \boldsymbol{\xi} \rangle + \kappa(\boldsymbol{\xi})\big) - \cos \kappa(\boldsymbol{\xi})}{|2\pi \boldsymbol{\xi}|},$$

we obtain

$$f(\mathbf{x}) - f(\mathbf{0}) = C_f \int_{\mathbb{R}^d} q_{\boldsymbol{\xi}}(\mathbf{x})\, d\Lambda(\boldsymbol{\xi}). \tag{5}$$

Since cos is 1-Lipschitz, $|q_{\boldsymbol{\xi}}(\mathbf{x})| \le |\langle \mathbf{x}, \boldsymbol{\xi} \rangle| / |\boldsymbol{\xi}| \le \|\mathbf{x}\| \le 1$ on the ball; $q_{\boldsymbol{\xi}}$ is continuous and depends on $\mathbf{x}$ only through the scalar $t = \langle \mathbf{x}, \boldsymbol{\xi}/|\boldsymbol{\xi}| \rangle \in [-1, 1]$.

Step 1 (convex combination). Partition $\mathbb{R}^d$ into cubes $I_{\boldsymbol{\theta}}$ of side $1/n$ centered at $\boldsymbol{\theta} \in \frac{1}{n}\mathbb{Z}^d$. The Riemann-type sum

$$C_f \sum_{\boldsymbol{\theta} \in \frac{1}{n}\mathbb{Z}^d} q_{\boldsymbol{\theta}}(\mathbf{x})\, \Lambda(I_{\boldsymbol{\theta}})$$

is a convex combination (the weights $\Lambda(I_{\boldsymbol{\theta}})$ are nonnegative and sum to 1) of the functions $C_f q_{\boldsymbol{\theta}}$, and it converges to the integral in (5) in $L_0^2(B_1^d)$ as $n \to \infty$. Hence $f - f(\mathbf{0})$ lies in the closed convex hull of $\{C_f q_{\boldsymbol{\xi}}\}$.

Step 2 (steps). Each $C_f q_{\boldsymbol{\xi}}$ is a function $r(t)$ of one variable on $[-1, 1]$ with $r(0) = 0$ and $|r'| \le C_f$, so its total variation is at most $2C_f < 2C$. Such a function is a uniform limit of staircase functions $\sum_k \beta_k \mathbf{1}_{\mathbb{R}_+}(\pm t - t_k)$ with $\sum_k |\beta_k| \le 2C$ (upward steps for $t > 0$, mirrored steps for $t < 0$). Each staircase is a convex combination of elements $\gamma\, \mathbf{1}_{\mathbb{R}_+}(\langle \mathbf{a}, \mathbf{x} \rangle + b)$ with $|\gamma| \le 2C$, so $C_f q_{\boldsymbol{\xi}} \in \overline{\mathrm{co}}(G_C)$. A closed convex hull of elements of $\overline{\mathrm{co}}(G_C)$ stays in $\overline{\mathrm{co}}(G_C)$, which proves the lemma. ∎

5.3 Completing the proof of Theorem 1

Proof of Barron's theorem. Let $f \in \Gamma_C$. By Lemma 8.16, $f|_{B_1^d} - f(\mathbf{0}) \in \overline{\mathrm{co}}(G_C)$, and every $g \in G_C$ has $\|g\|_{L_0^2(B_1^d)} \le 2C$. Lemma 8.15 with $H = L_0^2(B_1^d)$ and $B = 2C$ gives, for every $N \in \mathbb{N}$ and $c > 4C^2$, parameters $|\gamma_i| \le 2C$, $\mathbf{a}_i$, b_i with

$$\frac{1}{|B_1^d|}\int_{B_1^d}\Big|f(\mathbf{x}) - f(\mathbf{0}) - \frac{1}{N}\sum_{i=1}^{N}\gamma_i \mathbf{1}_{\mathbb{R}_+}\big(\langle \mathbf{a}_i, \mathbf{x}\rangle + b_i\big)\Big|^2 d\mathbf{x} < \frac{c}{N}.$$

Finally a sigmoid becomes a step under scaling: $\sigma(\lambda t) \to \mathbf{1}_{\mathbb{R}_+}(t)$ as $\lambda \to \infty$ for $t \neq 0$, and the hyperplanes $\{\langle \mathbf{a}_i, \mathbf{x}\rangle + b_i = 0\}$ have measure zero. By dominated convergence, replacing each indicator by $\sigma\big(\lambda(\langle \mathbf{a}_i, \mathbf{x}\rangle + b_i)\big)$ with λ large keeps the error below c/N. The result

$$f_N(\mathbf{x}) = f(\mathbf{0}) + \sum_{i=1}^{N}\frac{\gamma_i}{N}\,\sigma\big(\lambda\langle \mathbf{a}_i, \mathbf{x}\rangle + \lambda b_i\big)$$

is a one-hidden-layer network with N neurons; the constant $f(\mathbf{0})$ is the output bias. ∎

6 Generalization and Rademacher Complexity

6.1 Population and empirical risk

Let $\mathbf{x} \in \mathbb{R}^d$ be the input, $y \in \mathcal{Y}$ the output, and $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ iid training data from a distribution P. For a loss ℓ and a model class $\mathcal{F}$ we want

$$\arg\min_{f \in \mathcal{F}} \ \mathbb{E}_{(\mathbf{x},y)\sim P}\big[\ell(y, f(\mathbf{x}))\big] \quad \text{(population risk)},$$

but can only compute

$$\arg\min_{f \in \mathcal{F}} \ \frac{1}{n}\sum_{i=1}^{n}\ell(y_i, f(\mathbf{x}_i)) \quad \text{(empirical risk)}.$$

The generalization error of Section 1 is controlled by how far empirical averages can be from expectations uniformly over the class, $\sup_{f\in\mathcal{F}}\big|\frac{1}{n}\sum_i f(\mathbf{z}_i) - \mathbb{E}f\big|$, applied to the class of losses $\mathbf{z} = (\mathbf{x}, y) \mapsto \ell(y, f(\mathbf{x}))$.

6.2 Rademacher complexity

Definition 8.17: Rademacher variables and complexity

Random variables $\xi_1, \dots, \xi_n$ are **Rademacher** if they are iid with $P(\xi_i = 1) = P(\xi_i = -1) = \frac{1}{2}$. For a class $\mathcal{F}$ and iid samples $\mathbf{z}_1, \dots, \mathbf{z}_n$, the **Rademacher complexity** is

$$\mathrm{Rad}_n(\mathcal{F}) = \mathbb{E}\Big[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^{n}\xi_i f(\mathbf{z}_i)\Big],$$

the expectation being over both the ξ_i and the $\mathbf{z}_i$.

It measures how well the class can correlate with random signs: a rich class fits noise and has large complexity.

Theorem 8.18: uniform deviation bound

By symmetrization, $\mathbb{E}\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_i f(\mathbf{z}_i)-\mathbb{E}f\right| \le 2\,\mathrm{Rad}_n(\mathcal{F}\cup -\mathcal{F})$. If moreover every $f\in\mathcal{F}$ takes values in $[0,1]$, then with probability at least $1-\delta$

$$\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n} f(\mathbf{z}_i)-\mathbb{E}f\right| \le 2\,\mathrm{Rad}_n(\mathcal{F}\cup -\mathcal{F}) + \sqrt{\frac{\log(2/\delta)}{2n}}.$$

The lecture records this as "$\le 2\,\mathrm{Rad}(\mathcal{F}) + 1/\sqrt{2n}$", suppressing the confidence factor.

6.3 Two-layer networks

For $\mathcal{F} = \{f_{\boldsymbol{\theta}}(\mathbf{x}) = \sum_j \beta_j \sigma(\mathbf{w}_j^\top \mathbf{x})\}$ with $\sum_j |\beta_j| \le Q$, $\|\mathbf{w}_j\| \le W$, inputs $\|\mathbf{x}\| \le R$ and σ 1-Lipschitz with $\sigma(0)=0$ (e.g. tanh), the contraction lemma gives

$$\mathrm{Rad}_n(\mathcal{F}) \le \frac{2QWR}{\sqrt{n}} = O\Big(\frac{1}{\sqrt{n}}\Big).$$

The bound depends on the size Q of the outer weights, not on the number of neurons. Together with Barron's theorem, the approximation error decays like $1/N$ and the generalization error like $1/\sqrt{n}$, and neither rate depends on d.